Lesson 3.6 Guided Notes

Session 1 · Customary Unit Conversions

I can convert between customary units by multiplying or dividing by the conversion rate. · 6.RP.A.3d

Word / Spanish	Meaning — complete the blank	Example
customary units / unidades usuales	U.S. units like inches, _____, cups and gallons.	12 in = 1 ft
convert / convertir	Write the same amount with a _____ unit.	6 ft = 72 in
conversion factor / factor de conversión	A fact that shows two _____ amounts.	1 lb = 16 oz

Key idea

Big unit to small unit: _____. The number gets bigger.

Small unit to big unit: _____. The number gets smaller.

Worked example

A pitcher holds 2 quarts. How many fluid ounces is that? (1 qt = 4 cups, 1 cup = 8 fl oz)

Quarts to cups: 2 × 4 = _____ cups.

Cups to fluid ounces: 8 × 8 = _____ fl oz.

Check backward: 64 ÷ 8 = 8 cups, and 8 ÷ 4 = _____ quarts.

Answer: 2 quarts = _____ fluid ounces.

You try

1. Convert each measure.
a. 3 yd = _____ ft
b. 32 oz = _____ lb

_____

2. A banner needs 7 feet of fabric. Fabric is sold by the whole yard. How many yards should you buy? (1 yd = 3 ft)

_____

Sum it up: When I convert to a smaller unit, I ___ because ___.

_____

Session 2 · Metric Unit Conversions

I can convert between metric units by multiplying or dividing by powers of ten. · 6.RP.A.3d

Word / Spanish	Meaning — complete the blank	Example
metric units / unidades métricas	Meters, grams and liters, built on _____.	1 km = 1,000 m
kilo- / centi- / milli- / kilo- / centi- / mili-	Kilo means _____ times. Centi means one hundredth. Milli means one thousandth.	km, cm, mL
powers of 10 / potencias de 10	10, 100, 1,000. Metric changes _____ the decimal point.	3.5 m = 350 cm

Key idea

Big unit to small unit: _____ by 10, 100 or 1,000. Small to big: divide.

Make the units _____ before you divide or compare.

Worked example

A bag has 2.4 kg of flour. How many 300 g portions can you make? (1 kg = 1,000 g)

Make the units match: 2.4 × 1,000 = _____ g.

Divide: 2,400 ÷ 300 = _____.

Check: 8 × 300 = 2,400 g.

Answer: You can make _____ portions.

You try

1. Convert each measure.
a. 3.2 m = _____ cm
b. 750 g = _____ kg

_____

2. A tank holds 15 L of water. How many 750 mL jugs does it take to fill the tank? (1 L = 1,000 mL)

_____

Sum it up: To convert metric units, I move the decimal point because ___.

_____

Session 1 · Model, reason, and explain

Customary Unit Conversions · 6.RP.A.3d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A banner needs 8 feet of ribbon. Ribbon is sold only by the whole yard. Use 1 yard = 3 feet to find the exact length in yards and the purchase amount. Check whether 2 yards would be enough.

Yards	Feet
1	3
2	6
3	9
?	8

_____

_____

_____

Extend your reasoning

Explain why ounces of mass and fluid ounces of volume cannot be interchanged just because both use 'ounce'.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Metric Unit Conversions · 6.RP.A.3d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A container holds 1.5 liters. Cups hold 250 milliliters each. Complete the equivalent-measure table, find the number of cups, and explain why changing units does not change the amount of liquid.

Liters	Milliliters
1	1000
0.5	?
1.5	?

_____

_____

_____

Extend your reasoning

Explain 1.5 × 1000 by decomposing 1.5 into 1 + 0.5.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?